

MONTE-CARLO WORKFLOW · explicit RSA configuration**Simulation definition**

PackingDomain
Radius sampler
RSASOptions + seed
finite or periodic box

Python API

RSASimulator.run()
unit-aware inputs

C++ RSA engine

sample proposal
spatial-neighbour grid
overlap + periodic check
accept / reject / stop

PackingResult

centres + radii
class labels + ϕ
pair correlations $g_{ij}(r)$
visual diagnostics

ANALYTICAL WORKFLOW · Percus–Yevick reference**Mixture definition**

particle radii
number fractions
volume fraction
distance range

Python API

PercusYevickDomain
wavenumber = "auto"

C++ PY solver

mixture parameters
structure factors $S_{ij}(k)$
radial Fourier transform
grid-resolution warning

PY result

wavenumber grid
 $S_{ij}(k)$, $h_{ij}(r)$, $g_{ij}(r)$
mixture metadata
convergence diagnostics

Interpretation and validation

match composition • assess convergence • optional scattering